

KEEL v2.1

Adversarial Audit, Real Build Plan and Canonical Backend

Stress test | Product definition | Backend contracts | AI policy | Red team | Implementation gates

Verdict

BUILD THE PRODUCT. FREEZE THE BACKEND CONTRACTS.

The vision is now folded into a canonical backend plan: ownership, connections, ledger truth, recurring activity, paychecks, reimbursements, reconciliation, documents, notifications, investments, cards, governed AI, and repair operations. Shipping remains proof-gated.

Prepared	July 10, 2026
Basis	Complete KEEL archive + v2.1 backend canonicalization + adversarial workbook

How to use this document

This is not a summary of the uploaded archive. It is a zero-based decision document: what should be built, what should not be built yet, what would break under real financial data, and what must be true before KEEL can claim to be better than Quicken Classic, Monarch, Copilot, Simplifi, or AI-native accounting products.

The headline

The product is valid. The backend horizon is now explicit enough to support the intended end state, but production readiness is earned only through the canonical ledger, ownership, reconciliation, AI, notification, repair, export, and recovery gates.

Section	What it answers
1	Executive verdict
2	What the uploaded plan gets right
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14	Metrics, kill criteria, and first 90 days
15	Final decision
16	Canonical backend plan folded into scope
Appendices	Scope map, source notes, deliverables, and source-plan files

CAPABILITIES	P0 CANDIDATES	BACKEND GAPS	RED-TEAM ATTACKS
248	119	81	58

complete universe	must be narrowed by gates	production requirements	with kill criteria
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1. Executive verdict

Decision

Proceed with KEEL as a product thesis and prototype. Do not fund or schedule the uploaded v1.2 plan as written. Rename it from 'build-ready' to 'strategy-complete / architecture-incomplete,' then execute the staged plan in this report.

The archive did the hardest strategic work unusually well: it found a real seam between modern personal-finance apps and serious financial/accounting systems; it articulated deterministic financial truth beneath a calm interface; it recognized that personal, household, LLC, property, and investment activity must coexist without being conflated; and it treated AI as a proposal mechanism governed by evidence and approval rather than an oracle.

The plan fails at a different layer. It confuses an exhaustive feature taxonomy with an executable product sequence. Its final audit declares the system 'build-ready' and estimates roughly 27 agent-days after adding a multi-entity ledger, imports, reconciliation, receipts, portfolio lots, rental workflows, planning engines, tax packs, an MCP layer, responsive UX, and AI behavior. That is not aggressive; it is a category error. The estimate excludes the work that makes finance software trustworthy: provider failure handling, replay, statement evidence, journal revisions, permission tests, correction tooling, migration fixtures, support operations, calibration, security drills, and months of live-data exceptions.

The six changes that make the thesis investable

1. Narrow the primary buyer to an owner-operator household: a person with personal finances, one or more simple entities, and investments who is too complex for Monarch/Copilot but does not want a finance department.
2. Define the daily habit as a review-and-proof loop, not a dashboard and not a chatbot: one clear number, a small queue of consequential decisions, and a monthly close.
3. Use one truth model with two interaction modes: Focus when the user wants calm; Ledger when the user wants Quicken-grade detail.
4. Make trust the product moat: freshness, entity ownership, transfer integrity, reconciliation, source provenance, undo, export, and operator repair are user-visible features.
5. Treat AI-first as workflow intelligence - classification, matching, extraction, explanation, and typed command compilation - while disabling money movement, filing, payroll execution, trades, and personalized directives.
6. Replace the 27-day schedule with staged proof gates: 10-day interactive prototype, 6-8 week technical spine, 12-16 week private alpha, 6-9 month beta, 12-18 month category V1, and 24-36 month power parity.

Scorecard

Dimension	Score	Finding
Problem strength	9/10	The missing middle is real and painful for a paying prosumer.
Differentiation	8/10	Defendable if centered on entity-aware truth, migration, reconciliation, and proof.
Research quality	8/10	Broad and useful; some competitor assertions need recurring hands-on validation.
Feature completeness	9/10	The universe is excellent. It must be separated from launch scope.
Product coherence	7/10 -> 9/10	The Focus/Ledger model turns breadth into one coherent product.
Backend readiness	5/10	Core principles are right; production sync, revisions, operations, and evidence models are incomplete.
Execution realism	1/10 -> 7/10	The original estimate is unusable; the replacement roadmap is credible.
Trust/security readiness	5/10	Intent is strong, but design-partner launch needs concrete minimum controls and drills.
Business model	6/10	Potentially strong ARPU; free tier, aggregator cost, AI cost, and support burden need proof.

The product sentence

Recommended positioning

KEEL is the financial operating system for households with entities. Connect or import everything, keep personal and business boundaries clean, finish the few decisions that matter, and prove every number.

That sentence is materially better than 'the best of Quicken + Monarch + QuickBooks + Ramp + AI.' The latter describes a feature collage. The former describes a customer, a job, an operating model, and a reason to trust the result.

2. What the uploaded plan gets right

The archive is not a bad plan. It is a strong strategy document whose own final audit gave it more implementation certainty than it had earned. The following ideas should remain product law.

The Quicken-shaped hole

Quicken Classic remains unusually broad across personal cash flow, investments, business/rental activity, registers, reconciliation, and long history. Modern apps generally improve one surface while dropping depth. The archive correctly recognizes that this is not nostalgia; it is accumulated workflow coverage.

Entities before tags

Retrofitting ownership boundaries after launch is dangerous. A tag can describe a transaction, but it cannot by itself represent whose account it is, whether a transfer is an owner contribution, or how consolidated reports eliminate internal balances. Building plural entities and ownership into the spine is correct.

Deterministic spine, AI at fuzzy nodes

The archive's central architecture principle is exactly right. Financial math, posting, reconciliation, scopes, report semantics, and approval policy must be deterministic. AI is valuable where the world is ambiguous: merchant names, category/entity intent, receipt fields, match candidates, natural-language intent, and explanations.

Suggestion before action

The propose -> evidence -> approve -> execute -> audit pattern is the right default. It is especially important for entity assignment, transfers, receipt splits, rules, and any estimate that could affect tax or business understanding.

Migration as a first-class product

Power users do not switch because a landing page looks better. They switch when years of history, splits, categories, securities, transfers, and reconciled balances can move with quantified exceptions and rollback. The plan is right to make Quicken import a milestone rather than a support afterthought.

Protocol-shaped internal services

Designing internal contracts so they can later power an API/MCP surface is wise. It prevents the agent layer from becoming a privileged side door. The external protocol should arrive only after the same services have stable authorization, provenance, idempotency, and approval semantics.

Export and deletion as product promises

A financial system of record must let the user leave. Full export, document portability, and deletion workflows are not compliance footnotes; they are purchase criteria for sophisticated users.

What should be preserved verbatim as product law

- **No silent writes.** Material financial changes are proposed and approved; low-risk automatic actions remain reversible.
- **Aggregation is assumed to fail.** The user receives health state, repair tools, manual/import alternatives, and visible freshness.
- **Every number has a proof trail.** The product never hides scope, as-of time, exclusions, or formula version.
- **Every mutation is auditable and reversible.** An audit log alone is not undo; the ledger needs revisions, reversals, and period guards.
- **The user owns the data.** Export and deletion remain available even through downgrade or billing failure.
- **The app remains useful without AI.** AI accelerates and explains the financial system; it is not the financial system.

3. Zero-based market and product stress test

3.1 Does the problem justify a company?

Yes, with a narrower thesis than the archive occasionally implies. A person with W-2 income, investments, one or more LLCs, a property or project, receipts, owner transfers, and a household currently assembles a fragile stack: Quicken or spreadsheets for personal history; QuickBooks/Xero or ad hoc categories for entities; a receipt app; payroll software; a CPA handoff; and perhaps Monarch/Copilot for a modern view. The pain is not simply that the UI is ugly. The pain is that the same economic event has multiple meanings and nobody maintains the boundary cleanly.

The best initial customer is not a generic student, a 200-person business, or a complex family office. It is the owner-operator household: financially sophisticated enough to feel the seam, wealthy or busy enough to pay, but small enough that full accounting operations are excessive.

3.2 Why users might still refuse

- **Trust barrier.** A new company asks for the highest-sensitivity data a consumer owns. A beautiful prototype does not solve institutional trust.
- **Switching barrier.** Quicken users often have a decade or more of history, custom categories, investment transactions, reconciliations, and personal habits.
- **Connection barrier.** The product can be blamed for failures controlled by banks and aggregators. Reliability must be represented, measured, and repaired rather than marketed away.
- **Ambiguity barrier.** The same Home Depot purchase may be personal, a repair, an improvement, reimbursable, or mixed-use. AI does not remove the need for judgment.
- **Category gravity.** Monarch is simpler, QuickBooks is familiar to accountants, Quicken is deep, and Digits is increasingly AI-native. KEEL needs a job those products cannot satisfy through an easy extension.

3.3 The strongest argument against building KEEL

Bear case

The product sits in a commercially awkward seam. Users simple enough to avoid QuickBooks may be satisfied by Monarch plus tags; users complex enough to demand clean books may prefer QuickBooks plus a personal app. Connectivity and support costs are high, finance trust takes years, and incumbents can copy visible AI/receipt features faster than a startup can reproduce their data history and ecosystem.

3.4 Why the bear case is not fatal

The bear case is fatal only if KEEL is a broad dashboard. It is less fatal if KEEL owns a distinctive operating loop: explicit ownership, review, transfer and receipt matching, continuous reconciliation, entity-aware reporting, and proof. Those are not isolated UI features. Together they require a coherent data model and create switching costs through trusted history. An incumbent

can copy a receipt scanner or chat panel quickly; it cannot copy years of clean cross-entity state and user-specific decision memory without rebuilding its product center.

3.5 Necessary falsifiable hypotheses

Hypothesis	Falsifiable test
H1 - Paid pain	At least 15% of qualified owner-operator interviewees will pre-commit or pay \$25-\$75/month for a read-only/assisted design-partner product that demonstrably reduces financial cleanup.
H2 - Wedge advantage	Target users complete entity separation, owner transfer classification, monthly close, and CPA handoff materially faster and with higher confidence than in Monarch + spreadsheets or Quicken alone.
H3 - Daily habit	After onboarding, the median active user completes the review queue in under two minutes on at least three days per week, or completes a monthly close with repeated retention value.
H4 - Connection survivability	At least 90% of active design-partner account-days remain healthy or explicitly recoverable across the supported institution set.
H5 - Migration	At least 70% of qualified Quicken design partners can complete a verified migration or parallel validation without unquantified loss.
H6 - Economics	Paid cohort gross margin, including support labor, reaches a defensible target without hiding core value behind punitive connection/AI limits.

4. Competitive reality and the defensible wedge



Figure 1. Directional market map. The target is high financial depth without abandoning household calm. Scores are strategic, not contractual product claims.

The whitespace is narrower than the archive's final audit suggests. Current products are already moving across old category boundaries. Quicken's cloud Business & Personal product combines personal and business views and continues to improve. Monarch Plus adds business tracking, forecasting, and advanced investment features and uses multiple connectivity providers. Digits is not merely 'AI bookkeeping'; it now positions around an AI-native general ledger, reconciliation, close, invoicing, bill pay, reporting, mobile, and agent interfaces. QuickBooks continues to add AI-assisted categorization, reconciliation, receipt extraction, and financial Q&A. [S1-S6]

What each competitor is actually good at

Product	Core strength	Opening for KEEL
Quicken Classic B&P	Long-lived registers, reconciliation, investments, personal + business/rental workflows, reports and migration gravity.	Legacy interaction model, desktop center of gravity, cloud collaboration and modern AI expectations.
Quicken cloud B&P	Closest incumbent response to cross-device personal + business management.	Must be tested for depth, migration fidelity, reconciliation, investment detail and multi-entity boundaries.
Monarch Plus	Excellent household UX, sharing, net worth, planning, multiple aggregators, growing business and investment features.	Business remains lighter than accounting-grade entity books; investment transaction depth and proof workflows are not its center.

Product	Core strength	Opening for KEEL
Copilot Money	Fast, opinionated personal transaction UX and a strong daily review feel.	Primarily personal finance; business entities, accountant handoff, reconciliation and cross-platform depth are not the core.
Quicken Simplifi	Accessible cash-flow, spending plan and modern consumer experience.	Not the natural home for multi-entity, investment-accounting, or power-register needs.
QuickBooks Online	US accountant familiarity, business accounting depth, reports, receipts, payroll/integration ecosystem.	Not designed as a household, net-worth, retirement and personal planning command center.
Digits	The most important AI-native accounting threat: general ledger, close, reconciliation, Q&A and automation built around business books.	Business-first rather than whole-life; KEEL must not compete by merely adding an AI chat and P&L.
Ramp/Brex	Best-in-class receipt/spend automation because they control the transaction instrument and policy moment.	Corporate spend only; no personal/household/investment system of record.
Actual/Tiller	Control, portability, custom logic and power-user trust.	Require more user construction and do not deliver the full managed multi-entity, receipt and assistance loop.

The defensible wedge

Do not claim 'best at everything.'

Claim and prove this: KEEL is the best place to run a financially complex personal life with simple entities. It keeps ownership and transfers correct, closes the month, attaches the evidence, sees investments, and stays calm enough to use every day.

The wedge is an interlocking set of capabilities, not one checkbox: (1) account ownership and plural entities; (2) one review inbox across personal and business; (3) first-class transfers and due-to/from treatment; (4) continuous reconciliation and source proof; (5) receipt-to-transaction-to-tax-pack flow; (6) Quicken migration; and (7) natural language over deterministic data. Remove two or three of those and Monarch, Quicken, QuickBooks, or Digits becomes the obvious substitute.

5. The product KEEL should actually be

5.1 Primary persona

Owner-operator household

A person or couple with personal accounts, cards, 401(k)/IRA/brokerage assets, one to five simple entities or properties, receipts and tax categories, owner transfers, and professional handoff needs - but no desire to run a full finance department.

5.2 Jobs to be done

- **Know.** What is true right now across my life, which data is current, and which numbers are estimates?
- **Separate.** What belongs to me, my household, each LLC, each property, and each project?
- **Decide.** Which few ambiguous transactions, transfers, receipts, or anomalies need my judgment?
- **Prove.** Why is this number true, what source rows support it, and what changed since last month?
- **Close.** Are my accounts reconciled, receipts attached, exceptions resolved, and entity reports ready?
- **Hand off.** Can my spouse, bookkeeper, or CPA understand the output without inheriting my entire financial life?
- **Plan.** What is safely available after commitments, reserves, and clearly stated assumptions?

5.3 Product modes: Focus and Ledger

The product should not choose between Copilot-like calm and Quicken-like power. It should put them over the same financial truth model.

Mode	Experience	When it matters
Focus	One clear number, current-state confidence, three attention items, upcoming commitments, entity health, net worth and cash-flow snapshots.	Normal daily use, spouse/family visibility, mobile.
Ledger	Dense register, running balances, filters, splits, bulk edits, reconciliation, entity dimensions, statement proof, reports and audit history.	Power users, monthly close, migration, desktop.
Ask KEEL	Natural language compiles to a typed query or a proposed action, showing scope, formula, as-of time, rows and approval state.	Exploration and acceleration; never a privileged bypass.

5.4 The daily and monthly loops

1. Open KEEL and see the Clear number: liquid cash minus confirmed near-term commitments, configured reserves, and planned spend. Its proof is one click away.
2. Finish the Review inbox: high-impact and uncertain entity/category/transfer/receipt decisions first; low-value noise stays out.
3. Resolve connection or reconciliation exceptions only when the product can state exactly what is wrong and what repair will do.
4. At month end, complete the Close checklist by entity: accounts balanced, transfers matched, receipts/evidence status clear, exceptions disclosed, reports frozen or explicitly left open.
5. Export or share a scoped close packet with a professional. The professional can comment and request a correction without seeing unrelated personal data.

5.5 What KEEL is explicitly not at launch

- Not a bank, card, wallet, or money-movement product.
- Not payroll execution, tax filing, sales-tax compliance, inventory, or enterprise accounting.
- Not an investment adviser, trade execution tool, or personalized legal/tax directive engine.
- Not a generic AI chat wrapper and not a passive net-worth dashboard.
- Not 'free forever' live aggregation for unlimited accounts with unlimited support.

6. Responsive prototype and design system

A standalone interactive HTML prototype has been created alongside this report. It is intentionally not a static mood board. It demonstrates the information architecture, responsive behavior, review interaction, proof model, Focus/Ledger transition, receipt workflow, reconciliation workspace, entity reporting, investments, scenarios, trust center and an evidence-grounded Ask KEEL surface.

Design thesis

Focus when you want vibes. Ledger when you want Quicken. The user never changes products or data models; they change depth.

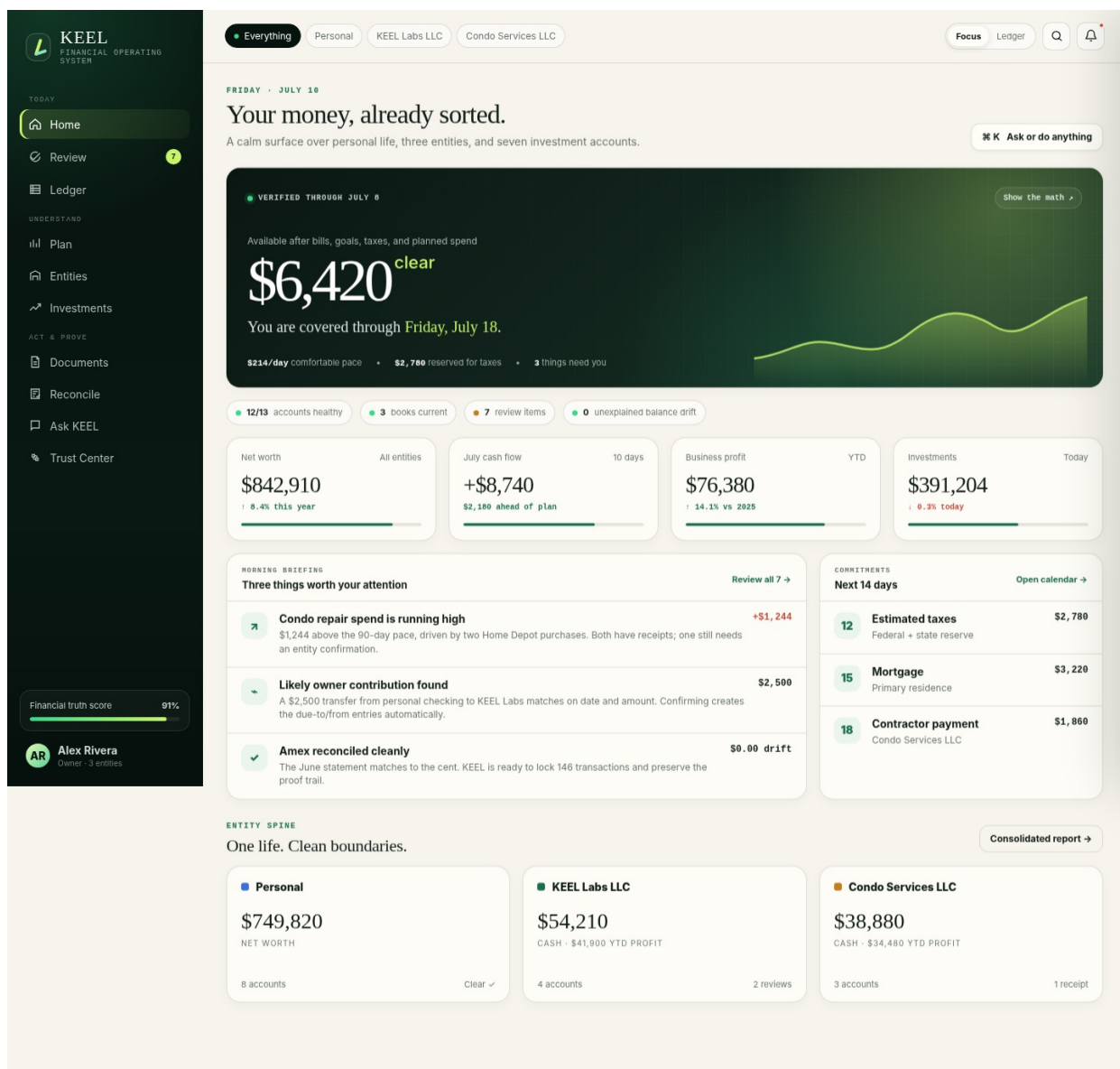


Figure 2. Desktop home: one clear number, visible truth state, commitments, entity spine and attention list.

The screenshot displays the KEEL v2 Review Inbox interface. The left sidebar contains navigation links: Home, Review (active), Ledger, Plan, Entities, Investments, Documents, Reconcile, Ask KEEL, and Trust Center. The main area shows a list of decisions with the following details:

- Decision 1:** Home Depot (HD), July 8 - Amex Business • 1002, -\$184.23. Confidence: 87%. Reason: Matched receipt, project "Unit 78 refresh," Schedule E candidate. Buttons: Not this, Edit, Approve.
- Decision 2:** Personal → KEEL Labs, July 7 - Same amount • 4 minutes apart, \$2,500.00. Confidence: 98%. Reason: Owner contribution. Exclude from income/spend; create due-to/from proof entries. Buttons: Not a transfer, Inspect, Confirm.
- Decision 3:** Amazon (a), July 6 - Chase Sapphire • 4419, -\$96.48. Confidence: 92%. Reason: Split into Office \$61.49 + Household \$34.99. Order detail found in connected email; two entities proposed. Buttons: Keep together, Edit split, Approve split.
- Decision 4:** Receipt from email (R), Ace Hardware - July 5, -\$72.19. Confidence: 81%. Reason: Match to pending Visa transaction. Amount exact; date differs by one day; merchant alias learned. Buttons: No match, View receipt, Match.

On the right, the **REASON TRACE** section explains why KEEL proposed these decisions, stating that every suggestion is reproducible. It lists four categories with their confidence levels:

- Account ownership:** Strong. Amex Business • 1002 has been assigned to Condo Services LLC since May 2025.
- Receipt evidence:** Strong. The receipt says "paint roller, grout repair, anchors" and totals \$184.23.
- Your prior corrections:** Medium. 11 of 13 Home Depot purchases on this card were categorized Repairs & maintenance.
- Tax mapping:** Needs judgment. Schedule E is suggested because the linked project belongs to Unit 7B. Confirm with your tax professional.

At the bottom of the reason trace, the **EVIDENCE** section lists: TXN-24819, DOC-1828, RULE-014, and 11 prior approvals. The bottom left shows a Financial truth score of 91% and the user Alex Rivera (Owner - 3 entities).

Figure 3. Review inbox: impact-ranked decisions, explicit confidence and a human-readable reason trace.

RECEIPT INTELLIGENCE

From paper to proof in one approval.

Capture → extract → match → itemize → entity → tax context. Every step stays reversible.

Forwarding address **Upload receipt**

Matched to Home Depot · -\$184.23 [Change](#)

Exact amount · same day · Amex #1002 · store geography consistent

ENTITY: Condo Services LLC PROJECT: Unit 7B refresh

TRANSACTION DATE: July 8, 2026 TAX CONTEXT: Schedule E candidate

Item	Category	Amount
Paint roller kit	Repairs & maintenance	\$38.97
Premixed grout	Repairs & maintenance	\$84.99
Wall anchors x 2	Supplies	\$36.98
Drop cloth	Repairs & maintenance	\$23.29
TOTAL		\$184.23

AT BOUNDARY
KEEL can organize this. It cannot decide tax treatment for you.
The Schedule E mapping is a suggestion based on the linked property and your prior approvals. The exported tax packet shows the evidence and confidence.

[Edit](#) [Approve & file](#)

Financial truth score: 91%
Alex Rivera
Owner · 3 entities

Figure 4. Receipt workspace: evidence, extraction, transaction match and line-item split without hiding uncertainty.

ASK KEEL

Answers with receipts.

Natural language compiles into deterministic queries. Every number links to the rows behind it.

TRY ASKING

- Why did Condo Services spend more this month?
- How much should I reserve for estimated taxes?
- Which business expenses are missing receipts?
- How long could I cover personal spending if income stopped?
- Which recurring charges increased?

KEEL Analyst
Read-only · cites every calculation

Why did Condo Services spend more this month?

ANSWER · BASED ON POSTED DATA THROUGH JULY 9

Repairs explain 78% of the increase.

Condo Services LLC spent \$6,284 in July so far, \$1,598 above the comparable first nine days of June. Two Unit 7B purchases and one contractor invoice account for most of the change.

Category	Amount
REPAIRS	+\$1,244
CONTRACTORS	+\$540
OTHER	-\$186

July delta = July 1-9 posted expenses (\$6,284) - June 1-9 posted expenses (\$4,686) = \$1,598

This is operational spend, not a new recurring baseline. The large items are linked to project **Unit 7B refresh**; one Home Depot transaction still needs your entity approval.

TXN-24619 · \$184.23 BILL-091 · \$1,860 TXN-24702 · \$1,609.77 [Report query](#)

Ask about cash, spend, taxes, entities, investments... [Ask](#)

Financial truth score: 91%
Alex Rivera
Owner · 3 entities

Figure 5. Ask KEEL: a financial answer with scope, as-of date, formula and linked source rows.

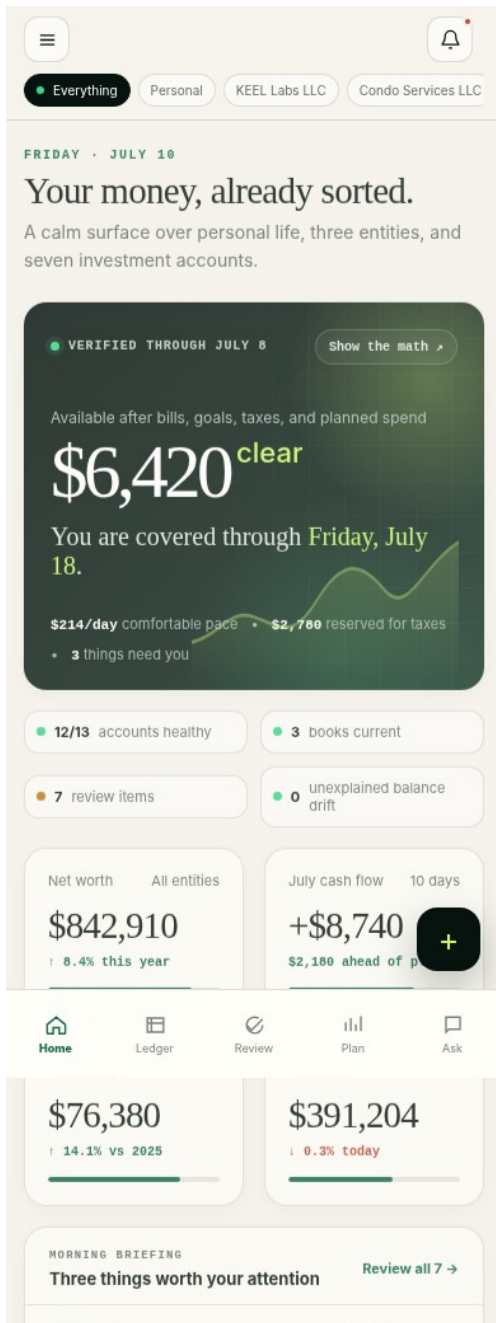


Figure 6. Mobile home: the calm daily loop survives at phone width; deep features remain accessible through the entity spine and bottom navigation.

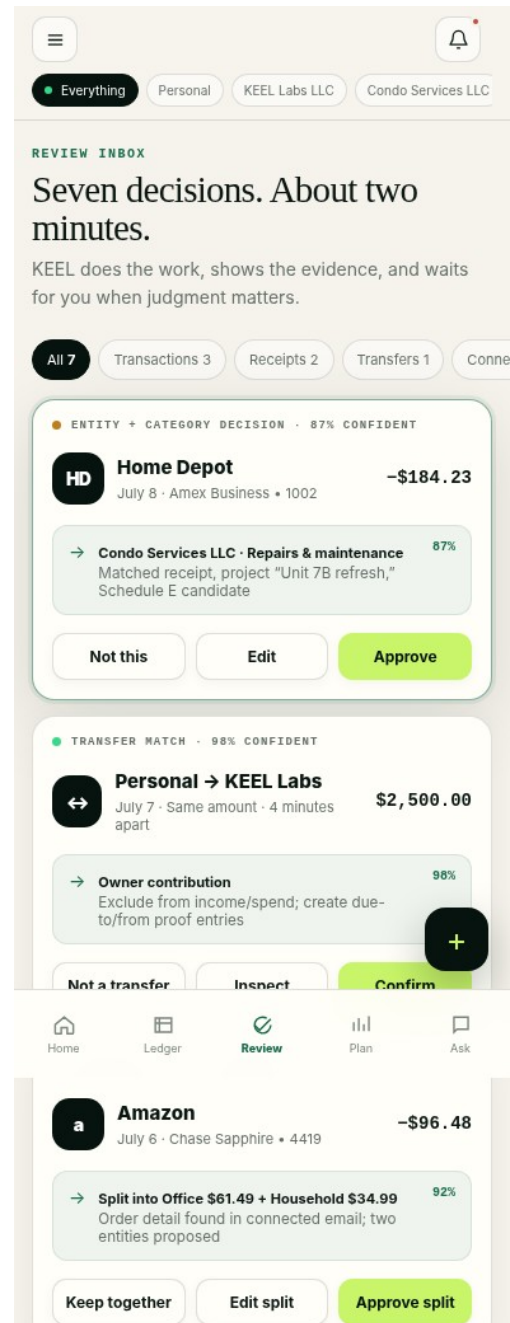


Figure 7. Mobile review: thumb-safe approvals with explicit alternatives and reversible decisions.

6.1 Visual principles

- **Financial calm, not fintech neon.** Warm paper, dark ink and restrained green create permanence; lime is reserved for decisive action and verified state.
- **Status before decoration.** Freshness, scope, reconciliation and source confidence are visually adjacent to the number they qualify.
- **Progressive disclosure.** A normal user sees meaning; a power user can reveal rows, formulas, journal consequences and audit history.

- **One attention grammar.** Cards consistently state the decision, consequence, evidence, confidence, alternatives and undo path.
- **Entity identity without fragmentation.** A persistent entity spine keeps personal, household and business scopes obvious while allowing consolidated views.
- **Mobile is a companion, not a shrunk desktop.** Home, Review, Plan, Ask and quick capture are first-class; dense reconciliation and migration remain desktop-optimized.

6.2 Prototype research script

1. Give a user three minutes with no explanation. Ask what KEEL does, what the main number means, and whether it looks current or estimated.
2. Ask the user to classify a Home Depot purchase, confirm an owner contribution, split an Amazon transaction, and match an emailed receipt. Observe whether evidence improves trust or slows the task.
3. Switch from Focus to Ledger and reconcile one account. Test whether the user understands that both modes share the same underlying truth.
4. Ask why Condo Services spent more this month. Verify that the user notices the as-of date, calculation, and cited rows before accepting the answer.
5. On mobile, complete the review queue one-handed and locate an entity-specific issue. Measure mis-taps, comprehension and time.
6. End by asking which existing tool the user would replace, which would remain, what data they would refuse to connect, and what proof they would need to pay.

7. Feature completeness without scope collapse

The companion workbook enumerates 248 detailed capabilities across connectivity/import, ledger truth, daily workflows, planning, entities/business, documents, investments, AI, security, operations and UX. That breadth is valuable as an architectural horizon. It is lethal as a simultaneous backlog.

UNIVERSE	P0 LABEL	PLAN MISSING	CRITICAL RISK
248	119	56	98
capabilities	requires gate reduction	explicit gaps	failure-sensitive

7.1 Replace feature tiers with three completeness tests

Test	Question
User-job completeness	Can a target user complete the whole job - import/connect, classify, reconcile, prove, close and export - without an external spreadsheet or support database edit?
Invariant completeness	Are ownership, balances, transfers, splits, periods, scopes, evidence and reports correct in success, failure, retry and undo states?
Failure completeness	Can the product represent stale, missing, duplicate, disputed, unsupported and partially complete data without silently presenting a false answer?

7.2 P0: what must exist for the first real alpha

- Provider-neutral connection contract, one live adapter, CSV/QIF/QFX staging, manual accounts, immutable raw events and explicit freshness.
- Accounts, ownership, plural entities, double-entry journal, opening balances, transaction lineage, splits, transfers, reversals and period guards.
- Review inbox, categories/entity proposals, rules, merchant aliases, bulk edits, universal search, Focus home and Ledger register.
- Balance snapshots, statement evidence, reconciliation sessions, drift diagnostics and calculation proof.
- Entity P&L and owner activity, consolidated view, export, deletion, scoped roles and an operator repair console.
- AI gateway with no-silent-write policy, prompt/model provenance, redaction, confidence calibration and a small exact-answer query grammar.
- Security minimum: strong auth, tenant isolation tests, secrets, encrypted documents, audit events, backup/restore and incident basics.

7.3 P1: what earns the trustworthy beta

- Receipt photo/email capture, extraction, matching, itemization, missing-receipt policy and proof packets.
- Safe-to-spend hardening, commitments, forecast uncertainty, goals, monthly close and grounded briefings.
- Second-provider strategy for the supported institution set, account lineage, better repair and status communication.
- Household/professional collaboration with adversarial scope tests.
- Grounded Ask KEEL over typed queries; saved queries and previewed write commands.
- Billing, entitlements, usage/COGS metering, privacy operations, security/recovery drills and mature support runbooks.

7.4 Deferred until evidence

Deferred area	Why
Investment lots and corporate actions	Holdings and allocation can arrive earlier. Tax-lot claims require transaction completeness, security normalization, brokerage cash and corporate-action fixtures.
Tax and retirement engines	Useful, but easy to overstate. Ship only with explicit assumptions, versioned math, benchmark fixtures and comprehension testing.
Invoicing, A/P and rental operations	These can pull KEEL into accounting verticals. Add modules only when retention/revenue evidence shows the owner-operator core is insufficient.
Payroll execution, filing and money movement	Integration/import only until there is a dedicated compliance, controls and partner strategy.
Personalized investment/tax/legal advice	Descriptive analytics and professional questions are enough initially. Regulated advice changes the company, not merely the feature list.
Retailer magic and consumer card policies	Potential delight later; not a prerequisite for proving the entity/reconciliation wedge.

7.5 Launch gates

Gate	Definition of done
Prototype	Users understand the one-number home, review queue, proof, Focus/Ledger relationship and phone behavior.
Technical spine	Balanced journal, raw replay, webhook idempotency, pending/posting lineage, undo, provenance and tenant isolation pass automated fixtures.
Private alpha	A user imports/connects, reviews, reconciles, reports, exports and repairs a month without hidden support actions.

Gate	Definition of done
Trustworthy beta	Receipts, grounded Q&A, collaboration, recovery drills, AI calibration, connectivity health and economics meet published internal thresholds.
Category V1	Verified Quicken migrations, entity/CPA packs, investments basic, open platform, trust posture and support SLOs pass.

8. Real build plan and staffing

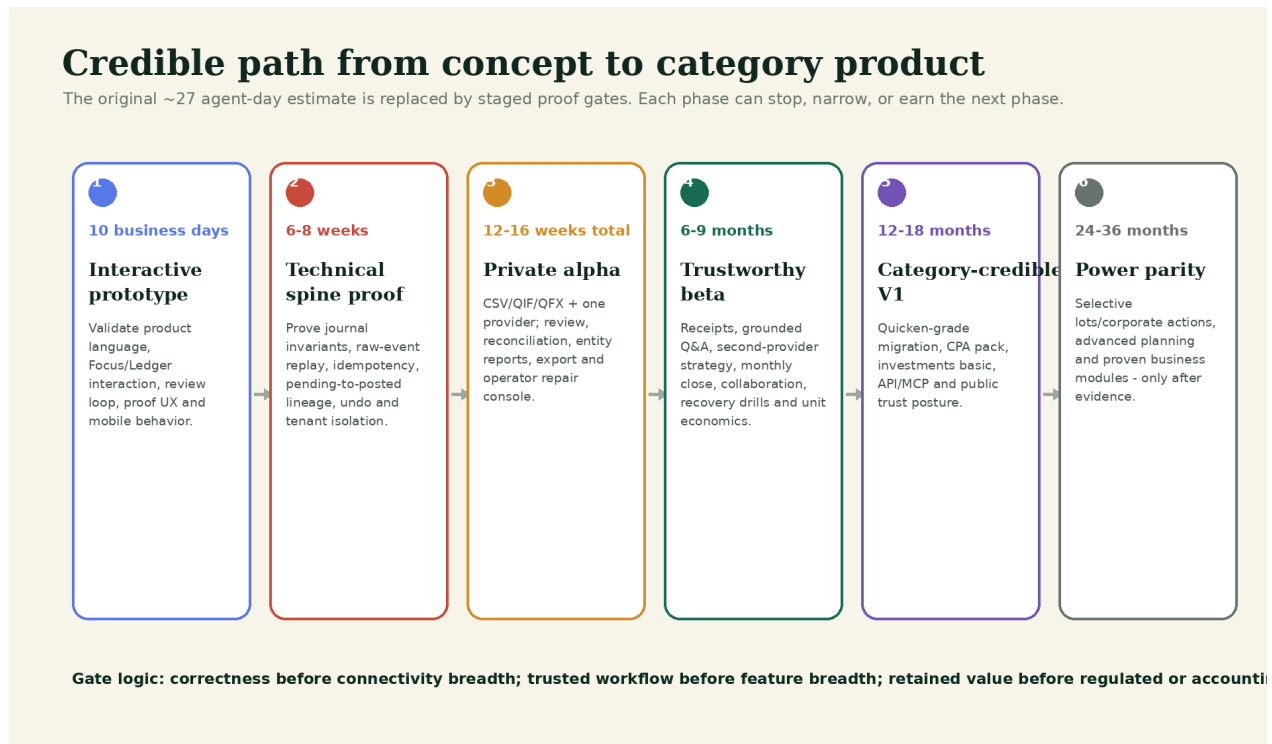


Figure 8. Proof-gated timeline. Dates are estimates for a focused, experienced team; live financial data will discover work no plan can enumerate in advance.

8.1 Stage 0 - 10 business-day interactive prototype

Timing	Deliverable
Days 1-2	Lock the owner-operator persona, three wedge jobs, product laws, non-goals, claims policy and test script.
Days 1-3	Build design tokens, responsive shell, entity spine, Focus/Ledger modes and stale/unknown/degraded states.
Days 2-5	Complete Home, Review, proof drawer, command palette and mobile daily loop.
Days 4-7	Complete Ledger, Reconcile, Receipts, Entities, Investments and Trust Center prototypes.
Days 6-9	Complete Ask KEEL, typed query preview, evidence citations and approval patterns.
Days 9-10	Run Quicken users and owner-operators through the prototype; revise positioning and flows before backend commitment.

8.2 Stage 1 - 6-8 week technical spine proof

This is not an MVP. It is a destructive test of the assumptions that would otherwise trigger a rewrite: can the system ingest unreliable events, maintain balanced and reversible state, isolate entities, reproduce calculations, and survive retry/replay? Use synthetic and anonymized fixtures before optimizing onboarding.

- Raw event store, provider-neutral adapter and sync simulator with duplicate, out-of-order, delayed, missing and changed events.
- Canonical accounts/transactions, entities/ownership, journal/postings, opening balances, splits, transfer links, revisions and period locks.
- Balance snapshots, statement lines, reconciliation and a query/proof service.
- CSV/QIF/QFX staging, Quicken dry-run report, exception inventory and batch rollback.
- Auth, RBAC/RLS, audit, tenant-isolation tests, encrypted documents/secrets, backup foundations and transactional outbox.
- AI gateway, typed policy matrix, provenance, redaction, baseline evaluation and approval-token contract.

8.3 Stage 2 - private alpha by week 12-16

The alpha supports manual/CSV/QIF/QFX and one carefully selected live provider. It should be read-mostly, explicitly bounded, and instrumented. The product is not ready because the screens work; it is ready when design partners can complete a monthly financial close without hidden database repair.

Alpha gate	Target
Activation	A qualified user reaches a verified home state in under 20 minutes or receives a precise blocker.
Review	Median queue time is under two minutes after the first week; high-confidence entity/transfer proposals meet thresholds.
Reconciliation	At least 80% of active accounts reconcile monthly without operator intervention.
Integrity	Confirmed internal transfers never affect income/spend; consolidated net worth does not change from an internal move.
Migration	Source counts, balances, splits and unsupported records are quantified before commit; import can be reversed.
Trust	Full export and deletion work; operator corrections are typed, reversible and audited.

8.4 Stage 3 - trustworthy beta in 6-9 months

Add receipts, stronger forecasting/commitments, grounded Q&A, collaboration, second-provider strategy, recovery drills, status/incident communication, billing/entitlements and cost/support measurement. Do not add advanced investment lots or broad business modules merely because the

core team is waiting on provider work; use that time to improve fixtures, support tooling and migration.

8.5 Stage 4 - category-credible V1 in 12-18 months

The public claim is earned when KEEL has verified Quicken migration, reliable entity close/report packets, basic investment visibility with completeness states, API/MCP contracts, robust exports, visible trust state, incident readiness, and acceptable gross margin including support. It does not require every item in the taxonomy.

8.6 Team

Role	Why it is required
Founder/product	Customer research, product law, prioritization, design-partner trust and fundraising/partnerships.
Staff backend/finance systems engineer	Ledger, revisions, reconciliation, proof, imports and financial invariants.
Data/integration engineer	Provider adapters, sync/replay, identity, matching, institution health and migration fixtures.
Senior frontend/product engineer	Responsive app, ledger performance, review loop, proof interaction and accessibility.
AI/product engineer	Extraction, classification, query compiler, evaluation, provenance and policy enforcement.
Product designer	Focus/Ledger system, onboarding, failure states, mobile and research.
Fractional finance domain lead	Controller/CPA review of reports, close, entity semantics and tax handoff boundaries.
Fractional security/privacy lead	Threat model, vendor review, minimum control program, incident and recovery exercises.

Reality check

A tiny team can build the prototype. A small senior team can prove the spine. A category-trustworthy product requires domain review, live-data operations, support tooling and time. Agents can accelerate code and fixtures; they do not compress provider behavior, user trust, security drills, or months of exception discovery into 27 days.

9. Backend and financial truth architecture

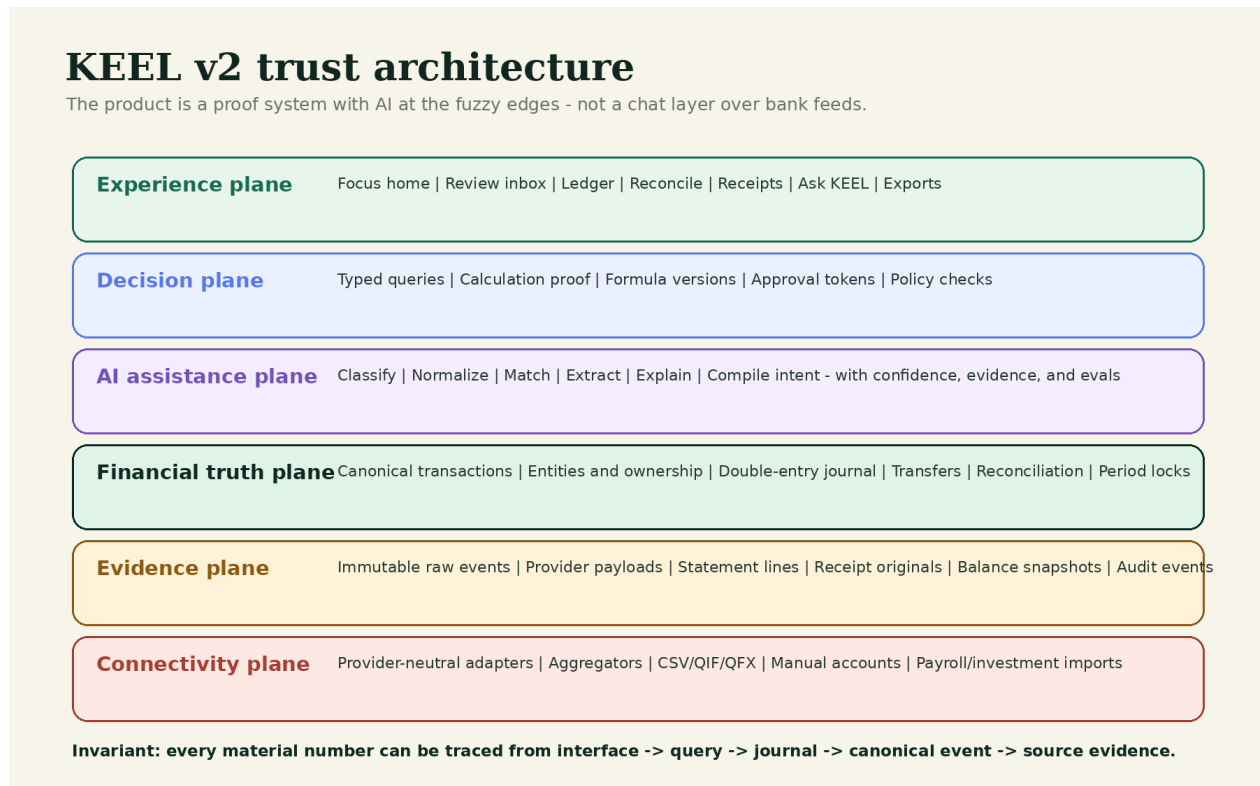


Figure 9. Recommended layered architecture. AI never writes around the truth, authorization, evidence or approval planes.

9.1 Seven non-negotiable invariants

Invariant	Acceptance
Balanced postings	Every committed posting batch balances by currency; reports derive from postings, not ad hoc category sums.
Source preservation	Provider/import/manual originals are immutable and independently addressable; normalization never destroys evidence.
Idempotent economics	Retry, replay and duplicate delivery cannot create a second economic event or posting.
Explicit ownership	Every account and reportable event has an ownership/entity scope or remains unresolved; inference is not silently treated as fact.
Reversible correction	Edits create revisions, reversals or compensating events; locked periods require explicit reopen with reason.
Scope-safe calculation	The same authorization compiler governs UI, reports, search, API, AI and MCP so hidden data cannot leak through aggregates.
Reproducible numbers	Every material result records as-of time, formula/query version, input scope, exclusions and source rows.

9.2 Ingestion and sync pipeline

1. Receive provider webhook or polling result into a minimal, hardened endpoint. Store an immutable raw event with idempotency key, provider cursor/version, received time, payload hash and connection scope.
2. Acknowledge quickly. Async workers validate ordering, deduplicate, normalize accounts/balances/transactions and preserve pending-to-posted or supersession lineage.
3. Write canonical changes and downstream tasks through a transactional outbox so a database commit cannot diverge from queue publication.
4. Run structural matching: duplicates, account identity, transfers, refunds and receipt candidates. Keep suggestions separate from approved economic state.
5. Promote approved or deterministic state into balanced journal postings. Reconciled/locked periods enforce mutation guards.
6. Update versioned materialized/report views and proof graph. Notify users only with scoped, actionable explanations.
7. Support can replay a connection in an isolated sandbox, compare raw/canonical/journal/report diffs, and promote a typed reversible repair - never hand-edit production rows.

9.3 Major backend gaps in the uploaded spec

Gap	Required correction
Raw events and replay	The DDL needs immutable payloads, cursor/checkpoint history, provider sequence semantics, replay runs and promotion diffs.
Pending/posting lineage	A transaction row needs supersession/lineage, source version and state history; simple pending/posted enums are insufficient.
Revisions and undo	Audit logs do not reverse books. Add journal/event revisions, void/reversal semantics, action bundles and locked-period guards.
Statement evidence	Reconciliation requires statements/sessions/lines, cleared state, differences, locks and independent proof.
Ownership and consolidation	Add account ownership, entity-specific mappings, due-to/from relationships, eliminations and consolidated report rules.
Identity graphs	Model merchants, vendors, customers and securities with source aliases and reversible canonical mapping.
Documents	Original hashes, versions, scan/quarantine, extraction provenance, many-to-many matches, line allocation and retention are required.
AI operations	Prompt/model/feature provenance, calibration, evaluation, policy, redaction, approval-token binding, rollback and canary cohorts are missing.

Gap	Required correction
Operator plane	Connection health, raw/canonical comparison, replay, correction events, support consent and status tooling must exist before external alpha.
Reliability/privacy	Coordinated DB/object backup, restore drill, data retention matrix, DSAR, incident/breach workflow and vendor program need owners and tests.

9.4 Core state machines

Object	State model
Connection	linking -> active -> degraded/reauth_required -> active; active -> disconnected. No sync promotion while authorization is invalid.
Transaction	pending -> posted -> reviewed -> locked; correction creates a new revision/supersession rather than rewriting the old event.
Transfer	suggested -> confirmed/rejected; cross-entity confirmation creates reciprocal due-to/from or explicit equity treatment after approval.
Document	uploaded -> quarantined/scanned -> extracted -> match_suggested -> matched/unmatched/disputed; financial splits require validated totals.
Reconciliation	open -> balanced -> locked; reopening requires role, reason and downstream invalidation.
Agent action	proposed -> approved -> executed/failed/rejected/expired; approval token binds the exact payload, actor, scope and version.
Import batch	uploaded -> parsed -> mapped -> dry_run -> committed/aborted -> reversed; exceptions and checksums survive every transition.

10. AI-first system, risk policy, and evaluation

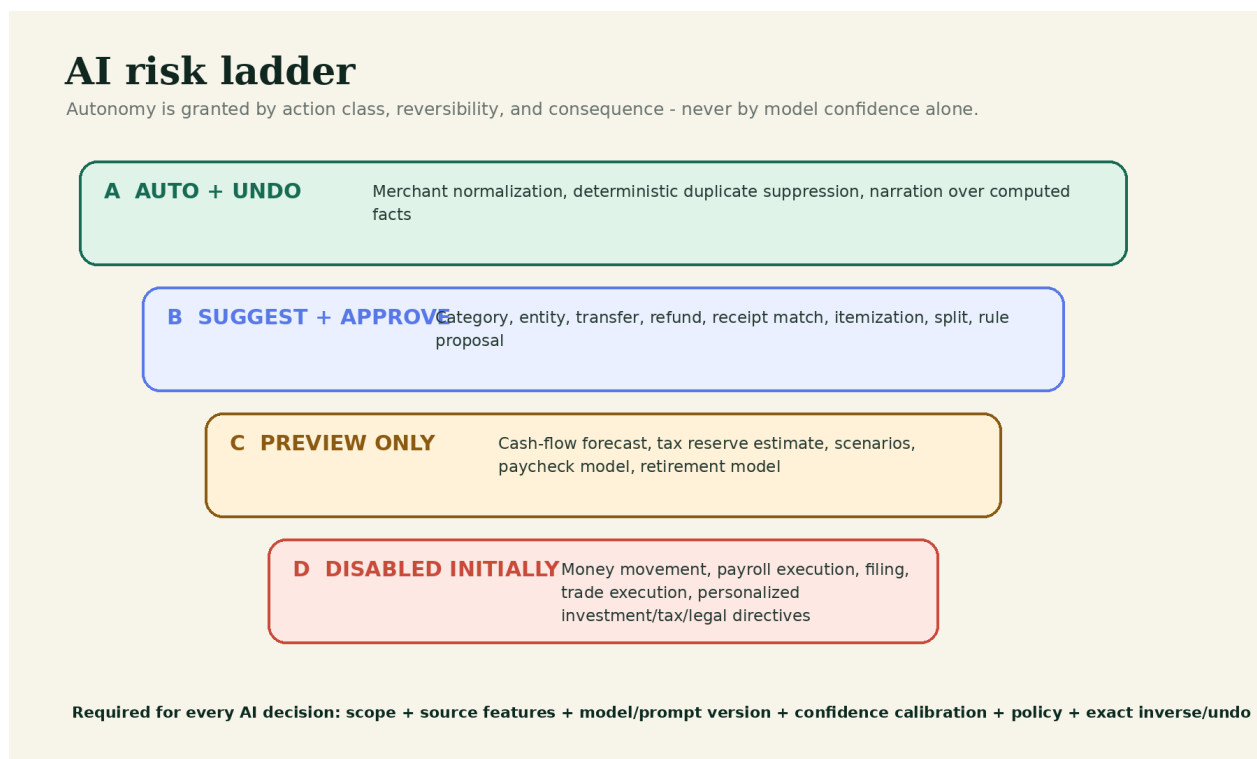


Figure 10. AI autonomy is determined by action class and consequence. Confidence changes routing within a class; it does not convert a prohibited action into an allowed one.

10.1 AI-first does not mean chat-first

The highest-value AI is mostly invisible. It cleans descriptors, proposes categories and entity ownership, detects transfers and duplicates, extracts receipts, matches evidence, prioritizes the review queue, explains changes and compiles natural language into deterministic queries or exact proposed actions. Chat is one interface to those services, not the product architecture.

10.2 Required decision record

- Task and policy class; user, household, entity and account scope.
- Source features and evidence references; sensitive fields excluded/redacted from model input.
- Model/provider/version, prompt/template/version, decoding configuration and output hash.
- Raw model proposal, normalized typed proposal and confidence/calibration bucket.
- Deterministic validators, conflicting rules, expected report/journal effect and required approval.
- User approval/rejection/correction, action payload hash, execution result and inverse/undo event.

10.3 Evaluation before launch

Task	Evaluation contract
Category classification	Precision/recall by category, user, merchant ambiguity and confidence bucket; cost-weight serious tax/entity errors more heavily.
Entity assignment	False-business and false-personal rates, cross-entity ambiguity, account ownership conflicts and evidence sufficiency.
Transfer/refund/duplicate matching	Pair precision, missed pair rate, amount/date variance and report-impact regression.
Receipt extraction	Field accuracy, totals, tax/tip, line items, bounding/source evidence and adversarial document text.
Grounded Q&A	Exact numeric answer, correct entity/date/account scope, as-of disclosure, formula and complete row citations.
Command compilation	Intent-to-typed-query accuracy, permission adherence, preview fidelity and exact approval-token execution.
Narration	No invented causes, no uncited material number, calibrated language for estimated/incomplete data.

10.4 Example interaction contract

User asks: 'Why did Condo Services spend more this month?'

KEEL compiles: entity = Condo Services LLC; metric = posted operating expenses; comparison = July 1-9 versus June 1-9; excluded = transfers, owner activity and pending transactions. The deterministic query returns +\$1,598. The model summarizes that repairs explain 78%, cites the three largest rows, states the data is posted through July 9, and offers to open the report. It does not calculate the number itself.

10.5 Advice boundary

At launch, KEEL should explain historical and projected data, show assumptions, identify questions, and prepare handoff. It should not tell a user to buy or sell a security, select an entity/tax election, file a return, set legally compliant payroll, or move money. The SEC treats robo-advisers as investment advisers with fiduciary obligations, and regulators have pursued misleading AI claims. The product copy and action policy must be reviewed as seriously as the model output. [S7-S8]

11. Trust, security, privacy, and regulatory boundaries

This section is a product and engineering control plan, not legal advice. Counsel should confirm applicability by business model, entity, data access method, geography and partnerships. The design target should nevertheless be finance-grade from the first external alpha.

11.1 Minimum external-alpha control set

Control area	Minimum
Identity	Passkeys or strong passwords, MFA, secure recovery, session management and step-up authentication for export, deletion and role changes.
Authorization	Central household/entity/resource policy, RLS as defense in depth, generated cross-tenant tests and scope-safe aggregate/query behavior.
Data protection	TLS, managed encryption, encrypted document storage, key/token rotation, least-privilege services and sensitive-log redaction.
Vendor management	Inventory and review of aggregators, AI providers, email/document services, cloud, analytics and support tools; contracts and exit plans.
Operations	Transactional outbox, idempotent jobs, rate limits, monitoring, data-quality console, audited corrections, status and incident communication.
Recovery	Coordinated database/object backups, point-in-time recovery, integrity hashes, RPO/RTO objectives and an actual restore exercise.
Privacy	Data map, retention matrix, user export/delete, DSAR workflow, household consent/revocation and support access controls.
AI governance	NIST-style govern/map/measure/manage loop, evaluation gates, model/prompt provenance, redaction and incident response for AI failures.

11.2 Current external environment

Open-banking policy remains in motion. The CFPB's Personal Financial Data Rights compliance page states that compliance dates were stayed after an October 2025 court order while reconsideration proceeds. KEEL should not base its first 18 months on a clean, universal open-banking future; it should architect for provider plurality, institution capability differences, user repair and high-quality import/manual paths. [S9]

The FTC Safeguards Rule and its breach-reporting amendment provide a practical security benchmark for businesses handling customer financial information, including required reporting to the FTC for certain breaches affecting 500 or more consumers. Exact applicability needs counsel, but designing below that standard would be commercially reckless for KEEL. [S10-S11]

For AI governance, the NIST AI Risk Management Framework's govern, map, measure and manage functions are a useful operating structure: establish policy and accountability; map tasks, users and consequences; measure accuracy, calibration, privacy and failure; then manage release, limits, monitoring and rollback. [S12]

11.3 User-visible trust center

- Connection health by institution/account: capability, provider, last successful step, balance/transaction as-of, known outage and recovery action.
- Book health by entity: accounts reconciled, unexplained drift, unresolved transfers, missing receipts, open period and report completeness.
- AI controls: action classes, current autonomy, provider/data-use disclosure, correction memory and deletion.
- Access: household/professional members, entity scopes, recent access, support consent and active API/MCP tokens.
- Data ownership: export, deletion, retention summary, backups/recovery statement and incident history/status.

12. Adversarial attack: how KEEL dies

The companion workbook contains 58 attacks with probability, severity, mitigations, proof tests, owners and kill/pivot conditions. The attacks below are the ones most likely to invalidate the company rather than merely create backlog.

Attack	Failure	Defense / decision
Feature-complete, habit-empty	KEEL ships many modules but no repeated behavior. Users admire the demo and keep using existing tools.	Make Review + Clear + monthly Close the only core activation/retention loops. Kill breadth that does not improve them.
Trust never crosses the link-account threshold	Prospects like the idea but will not connect accounts to a new company.	Read-only/import-first design partners, security package, export promise, founder credibility and progressive connection. Measure refusal reasons.
Monarch plus tags is good enough	The entity problem is not painful enough for most intended users.	Prove time/accuracy improvement on owner transfers, receipt evidence, reconciliation and CPA handoff. Pivot if no paid advantage.
QuickBooks is inevitable	Users who need clean entities eventually need payroll, sales tax, inventory, controls or accountant familiarity.	Own the personal/household side and excellent handoff/sync; exclude firms whose core job is operational accounting.
Connections break the promise	Provider failures create stale/duplicate/missing data and support floods.	Institution support tiers, provider-neutral design, explicit health, replay, fallback and excellent import/manual paths.
Migration betrays power users	A Quicken import loses splits, transfers, securities or reconciled balances without quantifying loss.	Dry run, golden corpus, balance proof, exception report, rollback and side-by-side period comparison.
The ledger cannot be repaired safely	Undo rewrites history, support edits SQL, or locked reports mutate.	Append-only revisions/reversals, typed correction events, period guards and operator console before external alpha.
AI makes a tax-consequential error	A personal expense is assigned to an LLC or an owner transfer becomes revenue.	Suggest/approve, evidence, weighted evals, unresolved states, explicit report impact and no legal/tax conclusion.
AI gives a convincing wrong answer	Chat scopes the wrong entity/date or invents a total.	Typed query compiler; LLM summarizes returned data; every material number is cited and as-of qualified.
Entity privacy leaks through aggregates	A hidden business/account is revealed through net worth, search or AI answer.	One authorization compiler across rows, aggregates, reports, search, exports, AI and MCP; adversarial inference tests.
Support economics erase margin	Every user has unique connection and history issues requiring founder intervention.	Support console, diagnostics, limited institution set, paid migration, runbooks and contribution margin including labor.
Free tier becomes a denial-of-wallet attack	Unlimited aggregation/AI/receipts produce cost without conversion.	Free demo/manual data; small live limit; meter actual COGS; preserve read/export access on downgrade.
Digits/Quicken/Monarch crosses the wedge	An incumbent adds enough cross-entity and AI capability to make KEEL redundant.	Build trusted history, migration, proof, personalized decision memory and open export - not merely visible features.
The team has no accounting adult	Subtle posting, ownership, close or report errors escape to users.	Fractional controller/CPA owns fixtures and sign-off; finance code paths require domain review.

Attack	Failure	Defense / decision
The schedule destroys the company	The team tries to build P0-P4 simultaneously based on the taxonomy.	Every task must map to a phase gate; no deferred module begins before core retention/correctness thresholds.

12.1 The most dangerous false-positive signals

- **Prototype praise.** People often praise financial dashboards they will not trust with data or pay to replace.
- **Connected-account count.** A user may connect accounts for novelty while retaining the old system of record.
- **AI question volume.** Chat novelty can be high even if answers do not change decisions or reduce cleanup.
- **Feature requests.** Prospects request payroll, crypto, credit, rental, invoices and tax because the product concept is broad. Requests are not evidence of a coherent paid wedge.
- **Categorization accuracy alone.** A 95% classifier can still fail catastrophically on the 5% that affects entities, transfers, tax and reconciled reports.
- **Gross software margin excluding support.** A finance app can look healthy until connection/migration support labor is allocated to cohorts.

12.2 Kill and pivot conditions

Condition	Trigger
Paid demand	Fewer than 10% of repeatedly qualified prospects will pay at least \$25/month after a tested, credible prototype and concierge workflow.
Wedge efficacy	Target users show no material improvement over Monarch/Quicken/QuickBooks combinations on entity separation, close or handoff.
Connectivity	Supported design-partner institutions remain healthy/recoverable on fewer than 90% of account-days after focused provider work.
Support burden	Mature paid cohorts require more than roughly 30 minutes of human data repair/support per user per month with no declining trend.
Migration	Fewer than 70% of qualified Quicken users can complete a verified import/parallel run within the declared coverage boundary.
Habit	Fewer than 25% of weekly active users complete review or close actions; usage is predominantly passive balance checking.
Economics	Contribution margin remains negative at a price users accept after connection, AI, storage, receipt and support costs.

13. Business model, economics, and go-to-market

13.1 Pricing hypothesis

The uploaded Free / Plus \$99 / Pro \$249 structure is attractive but likely too cheap for the heaviest owner-operator users if live connections, receipt extraction, AI questions and high-touch support are included. Pricing should be tested as a willingness-to-pay and cost-control instrument rather than copied from consumer budgeting apps.

Tier	Hypothesis	Purpose
Free / demo	Sample data, manual account or limited import, one planning surface, limited AI demonstration, full export of user-entered data.	Acquire and teach without open-ended aggregator/support cost.
Personal	Approximately \$12-\$20/month or annual equivalent: household, unlimited manual data, reasonable live accounts, review, budgets, net worth, basic investments and Ask limits.	Compete with premium personal tools while preserving margin.
Owner	Approximately \$29-\$59/month: multiple entities, receipt workflow, reconciliation, entity reports, tax reserve, Quicken migration tools and professional access.	Primary wedge and likely best product-market fit.
Pro / reviewed	Approximately \$99-\$299/month depending on human service: prioritized migration, close review, bookkeeper/CPA workflow, higher limits and support SLA.	Hybrid software + expert layer; do not imply accounting/tax representation without the service/legal structure.

13.2 Cost model that must be measured

- Per connected item/account/provider and refresh/backfill behavior; institution-specific reconnect support.
- AI tokens/calls by task, especially receipt extraction and long-context Q&A; cache and deterministic pre-processing opportunities.
- Document storage, malware scanning, OCR/extraction, email ingestion and export bandwidth.
- Market/investment data and identifier normalization if holdings expand.
- Support and migration minutes by user, provider, import source and product tier.
- Security/compliance/vendor review, observability and incident costs that do not appear in a per-request model.

13.3 Go-to-market wedge

Acquisition thesis

Migration is not a utility after signup. It is the campaign: 'Keep the power and history of Quicken. Lose the desktop-era workflow. Add clean entities, receipts and proof.'

- **Design-partner cohort.** Recruit 15-25 owner-operators with Quicken, multiple entities/properties and a real month-end cleanup process. Charge something; unpaid enthusiasm is weak evidence.
- **Quicken migration clinic.** Offer a dry-run report before requiring a switch. Quantify accounts, history, transfers, splits, categories, securities, balances and exceptions.
- **CPA/bookkeeper channel.** Do not ask professionals to abandon QuickBooks. Give them a cleaner scoped packet and unresolved-item workflow for clients whose personal and entity lives are entangled.
- **Founder/creator/consultant communities.** Target people with W-2 + 1099/LLC/investment complexity. Use before/after monthly-close stories, not generic AI claims.
- **Trust content.** Publish connection status philosophy, calculation proofs, export format, model/action policy and migration limits. The willingness to state limitations is differentiation.

13.4 Messaging hierarchy

Layer	Message
Hero	Your money, already sorted.
Subhead	One calm view across personal life, entities and investments - with proof behind every number.
Primary proof	Finish the seven decisions that matter, reconcile the month, and hand off clean entity records.
Power proof	Focus when you want clarity. Ledger when you want Quicken.
AI proof	AI finds, matches and explains. You approve what changes the books.
Trust proof	See freshness, source, formula, scope and undo. Export everything whenever you want.

14. Metrics, kill criteria, and first 90 days

14.1 Product and trust metrics

Metric family	What to measure
Activation	Connected/imported accounts with explicit ownership; first review complete; first proof viewed; verified home reached.
Habit	Review days per active week; median queue duration; monthly close completion; unresolved item age.
Integrity	Transfer false positives/negatives; entity correction rate; reconciliation success; unexplained drift; report restatements.
Connectivity	Healthy account-days, freshness, reauth rate, duplicate rate, pending/posting repair rate, provider/institution support incidents.
AI	Precision by task and confidence bucket; approval/correction rate; uncited-number rate; scope error; eval regression and policy violations.
Migration	Dry-run completion, quantified exception rate, balance match, rollback, verified switch/parallel-run conversion.
Trust	Export usage, proof engagement, trust-center visits, connection refusal reason, security/support incidents and deletion completion.
Economics	Paid conversion, ARPU, gross margin, contribution margin including support, support minutes, AI/connection/document cost per cohort.

14.2 First 90 days

Window	Work	Exit evidence
Days 1-10	Complete and test the interactive prototype; lock the product charter, non-goals, design system, research script and claims matrix.	Prototype comprehension and willingness-to-pay evidence.
Days 11-20	Recruit and interview 20-30 qualified owner-operators; collect anonymized Quicken/import fixtures and monthly-close workflows.	At least five paid/committed design partners and a ranked problem corpus.
Days 15-30	Write architecture decisions, invariant/property tests, canonical schemas, provider adapter contract, threat model and minimum security program.	Spine backlog tied to launch gates rather than feature tiers.

Window	Work	Exit evidence
Days 21-45	Build raw events, sync simulator, journal/entities, revisions, import staging, basic reconciliation and tenant-isolation harness.	Duplicate/out-of-order/pending-posted/undo fixtures pass.
Days 35-55	Build responsive shell, Focus home, Review, Ledger, proof drawer and operator diagnostics against synthetic/import data.	One complete manual/import month can be reviewed and proved.
Days 45-65	Build Quicken dry run, balance/exception report, batch rollback and first migration clinic with design partners.	Quantified migration results; no silent loss.
Days 55-75	Add one live provider in a controlled cohort, connection health, replay sandbox and daily SLO measurement.	Four-week health experiment begins.
Days 65-85	Add reconciliation sessions, entity P&L/owner activity, export and typed correction events; run close with partners.	Users close a month without direct DB repair.
Days 80-90	Review paid demand, wedge performance, integrity, connectivity, support and team velocity. Decide alpha continuation, narrower pivot or stop.	Written gate decision with evidence and revised budget.

14.3 Founder decisions required now

1. Commit to the primary persona and say no to generic students, scaled SMBs and complex accounting as the launch market.
2. Select a product name/domain only after positioning tests; do not let brand work delay the product charter or design-partner recruitment.
3. Recruit a finance-systems engineer and a fractional controller/CPA before treating the schema as authoritative.
4. Choose the first import sources and live institution/provider cohort based on design-partner reality, not market-size theater.
5. Set a security/trust minimum for any external data; no 'private beta' exception to tenant isolation, secrets, backups and incident basics.
6. Establish the gate review cadence and empower it to delete features, not merely reschedule them.

15. Final decision

Go - with conditions

KEEL is a valid product and a credible company thesis. The missing product is not an omnipotent finance app. It is a calm, entity-aware system of financial truth for owner-operator households. Build the prototype immediately; build the financial spine only against explicit proof gates; expand only when retained paid behavior earns it.

The archive's ambition should remain. Its claim of completeness should be reframed. A product can carry a complete architectural horizon without implementing every module. The goal is not to make every competitor irrelevant on day one. The goal is to make the target user unwilling to return to a stack where personal and business boundaries, receipts, transfers, reconciliations and explanations live in separate products.

KEEL wins when a user can open one calm surface, know what is true, finish the seven decisions that matter, click any number to prove it, close each entity cleanly, and preserve enough depth that the Quicken power user does not feel infantilized. The prototype demonstrates that this can feel modern. The revised architecture and roadmap demonstrate that it can be built responsibly. The red team defines the evidence required to continue.

Non-negotiable founder mantra

One life. Clean boundaries. Provable numbers. AI does the ambiguity work; the user retains authority; the ledger retains truth.

16. Canonical backend plan folded into scope

This section is the implementation-level answer to the final backend review. It is folded into the source corpus as **15-KEEL-BACKEND-CANONICALIZATION-v2.1.md**. Where earlier schema examples are simpler, this section and doc 15 control.

16.1 Final scope answer

YES: the architecture now covers the backend needed for Quicken-grade account management and reconciliation, Monarch/Copilot-grade personal finance and sharing, multi-entity business-lite, receipts, recurring expenses, structured paychecks, friend/employer reimbursements, investments, notifications, card intelligence, and governed AI.

NO: that does not mean every specialist module ships in the first release. Completeness means later modules attach to the same durable truth model rather than forcing a rewrite.

16.2 Canonical domains

Domain	Mandatory backend contract	What it unlocks	Gate
Identity & sharing	Multi-household memberships, entity memberships, joint account owners, credential ownership, professional links, resource permissions	Private/shared accounts, family collaboration, CPA portfolios, correct offboarding	No scope leakage through API, report, search, AI, export, notification
Connections & ingestion	Provider-neutral adapters, immutable raw events, idempotency, checkpoints/replay, balance snapshots, account lineage, import batches	Reliable sync, fallback providers, replacement-card history, Quicken migration	Duplicate/out-of-order/reconnect/import fixtures yield one economic history
Ledger & classification	Canonical transactions, balanced journal batches/postings, revisions, category-to-account mappings, dimensions, split templates, rule versions	Nested categories, trips/projects/properties, business reports, universal undo	Balance, split conservation, revision, and locked-period property tests
Recurring	Series, patterns, occurrences, candidates, predictions, overrides, status events	Bills, subscriptions, paydays, variable utilities, skips, pauses, cancellations, price changes	Backtest fixed, variable, skipped, resumed, and cancelled series
Paychecks	Typed components, templates, source fields, payroll imports, many-to-many destination matches	Gross-to-net, taxes, benefits, 401(k), HSA, equity, reimbursements, split direct deposit	Components reconcile to gross, net, and matched accounts
Transfers & settlements	Transfer links, counterparties, expense shares, claims, settlements, refund expectations/matches	Card payments, Venmo/Zelle friend payments, employer/client reimbursements, owner movement	Internal transfers never affect income/spend; claims settle correctly

Domain	Mandatory backend contract	What it unlocks	Gate
Reconciliation & close	Statements, statement lines, sessions, items, adjustments, balance snapshots, period locks, close checklists	Quicken-style statement proof, explained drift, entity close	Every difference explained; locked history rejects mutation
Documents & receipts	Immutable document versions/hashes, extracted fields, line items, many-to-many matches, evidence policies	OCR, itemization, missing receipts, tax/CPA proof packets	Source-to-extraction-to-posting proof and exact totals
Notifications	Domain events, rules, preferences, deliveries/history, snoozes, escalations, push subscriptions	In-app/email/web/native/SMS attention without noisy feature-specific code	Dedupe, retry, quiet hours, permissions, privacy, deep links
AI & agents	Typed answers, evidence refs, proposals, approval tokens, memories, model/prompt/policy versions, eval runs	Concise TLDRs, grounded questions, safe classification/matching/extraction and proposed actions	Exact-answer, calibration, injection, replay, and no-silent-write tests
Investments	Security aliases, holdings, transactions, portfolio cash, lots, corporate actions, valuation snapshots, completeness flags	401(k)/brokerage visibility through later lot and performance depth	Unsupported precision is disabled when data is incomplete
Cards & offers	Card catalog, reward rules/caps/benefits, merchant categories, point valuations, recommendations; separate offer eligibility/redemption	Which-card-to-use guidance, utilization/APR/benefit insights, later partner offers	Affiliate availability cannot change unbiased ranking
Support & recovery	Typed inspect/replay/merge/split/reverse/recompute/correct actions, JIT access, coordinated backups	Finance-grade support without hidden SQL edits; reproducible incidents and recovery	Synthetic failures repaired via APIs; export/restore reproduce cited evidence

16.3 Category, split, recurring, paycheck, and Venmo depth

- Categories may nest deeply - for example Travel > Vacation > Japan 2027 > Lodging - while trip, project, property, client, person, tax context, and reimbursable party remain independent dimensions.
- A transaction can split by dollar amount or percentage, across categories, entities, projects, people, or periods, with reusable templates and exact conservation.
- Recurring records represent expected occurrences and actual matches, not merely a next-due date. They support variable ranges, skipped months, pauses, cancellation, resumption, price increases, and recurring split/entity rules.
- A paycheck is a set of typed components. One pay event may match multiple cash deposits, a 401(k), HSA, brokerage/ESPP transaction, employer match, tax withholding, and reimbursement.
- A Venmo/Zelle/Cash App payment from a friend settles a reimbursement claim. It is not income. The original shared expense remains historically accurate while the user's economic share is reported correctly.

16.4 Structured AI contract

Material AI outputs are typed records, not free-form authority. The default response exposes a concise answer first and proof on demand.

Field	Requirement
verdict	yes, no, or uncertain
tldr	One concise answer suitable for the default UI and notification
confidence	Empirically calibrated by task, not decorative
as_of and scope	Time, entities, accounts, period, and relevant completeness/freshness
reason_codes	Machine-readable explanation categories
evidence_refs	Transactions, statements, receipts, recurring series, postings, or calculation records
proposed_actions	Typed changes that can be previewed and edited
requires_approval	Server-enforced policy result; approval token binds exact payload, actor, scope, version, and expiry

AI may normalize merchants, propose categories/entities/transfers/refunds/reimbursements/duplicates, extract documents, explain deterministic anomalies, compile natural language into typed queries/actions, and prepare concise briefings.

AI may not perform ledger arithmetic, bypass permissions, silently change material records, claim incomplete data is reconciled, move money, run payroll, file taxes, trade, or provide regulated directives in the initial product.

16.5 Shared service surface

The web app, mobile client, public API, MCP server, and support console must call the same authorized domain contracts:

```
accounts.* connections.* members.* permissions.* entities.*
transactions.* categories.* dimensions.* rules.* recurring.*
transfers.* counterparties.* reimbursements.* refunds.*
statements.* reconciliations.* close.* documents.* receipts.*
paychecks.* income.* budgets.* goals.* forecasts.* scenarios.* reports.*
investments.* securities.* lots.* cards.* rewards.* offers.*
notifications.* preferences.* ai.* knowledge.* proposals.* approvals.* provenance.*
imports.* exports.* audit.* support.*
```

16.6 Core now, additive later, explicitly deferred

Scope	Included
Core financial foundation	Ownership/permissions; connections/imports/account lineage; ledger/categories/dimensions/splits/rules; transfers/reimbursements; recurring/paychecks; reconciliation/close; budgets/goals/cash flow/net worth/reports; receipts; notifications; governed AI; exports/repair/recovery
Additive later modules	Advanced lots/corporate actions/equity/crypto; card

Scope	Included
	rewards/offers; retailer itemization; cancellation/negotiation; property/lease operations; tax/retirement scenarios; QuickBooks/Xero and payroll integrations; native apps/widgets/watch/voice
Explicitly deferred regulated scope	Money movement and bill pay; payroll execution; tax filing; securities trading or personalized investment advice; card issuing; inventory, sales-tax engine, and enterprise ERP

16.7 Final implementation gates

1. Ledger: every batch balances by currency; splits conserve; revisions and period locks work.
2. Ingestion: duplicate/out-of-order events, reconnects, imports, and pending-to-posted changes replay deterministically.
3. Ownership: no resource or aggregate leaks through any user, professional, API, AI, notification, export, or support surface.
4. Relationships: transfers, owner movements, friend/employer reimbursements, refunds, and settlements report correct economic meaning.
5. Recurring/paycheck: expected occurrences backtest; paycheck components reconcile to gross, net, and destinations.
6. Reconciliation: statements and cleared ledger activity close with every difference explained.
7. Documents: extraction and matching retain exact source proof and amount validation.
8. AI: exact-answer, confidence calibration, evidence completeness, prompt-injection, approval replay, and no-silent-write gates pass.
9. Notifications: preferences, privacy, dedupe, retry, delivery history, and deep links work.
10. Operations: common data failures are repaired through typed audited APIs; full export and isolated restore reproduce cited evidence.

16.8 Concise founder TLDR

YES - the plan now contains the backend and API horizon needed for the intended best-in-class product.

NO - this is not permission to build every module simultaneously or call a screen a finished feature.

WHY - the durable financial spine is now explicit, so later UI and specialist modules can be added without changing the meaning, ownership, proof, or history of the money.

Appendix A - Revised scope map

Scope	Capabilities
Build first	Entities/ownership, import/migration, journal/revisions, transaction lineage, review, transfers, reconciliation, proof, export, operator repair, security minimum.

Scope	Capabilities
Build for beta	Receipts, monthly close, safe-to-spend/commitments, grounded Q&A, collaboration, second-provider strategy, billing/economics.
Build for V1	Quicken migration breadth, CPA packet, investments basic/completeness, API/MCP, public trust posture.
Defer	Lots/corporate actions, retirement/tax scenario suite, property operations, invoices/AP, payroll helpers, credit/crypto/retailer magic.
Do not build initially	Money movement, payroll execution, filing, inventory, sales tax, trade execution, personalized regulated advice.

Appendix B - Companion deliverables

Artifact	Filename	Contents
Interactive prototype	KEEL_v2_interactive_prototype.zip (experience prototype; unchanged)	Standalone responsive HTML; no backend or external dependencies. Includes six QA screenshots and README.
Completeness workbook	KEEL_v2.1_feature_completeness_and_read_team.xlsx	248 capabilities, launch gates, 81 backend/trust requirements, 58 attacks, canonical backend sheet, competitor matrix, roadmap, and sources.
This report	KEEL_v2.1_Adversarial_Audit_and_Build_Plan.docx / .pdf	Founder-ready decision document plus the canonical backend scope, AI response contract, and implementation gates.
Complete source plan	KEEL_v2.1_complete_plan.zip	Updated Markdown corpus, canonical backend patch, changelog, report, workbook, and supporting prototype.

Appendix C - Source notes

Current product and regulatory assertions in this report are intentionally based primarily on official product pages, official documentation and regulator sources. The competitive scores in the workbook are directional and should be revalidated through paid hands-on trials before external claims or investment decisions.

ID	Source
S1	Quicken Business & Personal product page
S2	Quicken Classic Business & Personal product page
S3	Monarch product/help documentation, including Plus and connectivity
S4	Digits product and feature documentation
S5	QuickBooks AI accounting
S6	Plaid documentation: transactions, webhooks, institution status and balances
S7	SEC robo-adviser guidance / fiduciary context
S8	SEC enforcement and investor communications on misleading AI claims
S9	CFPB Personal Financial Data Rights compliance resources
S10	FTC Gramm-Leach-Bliley Act / Safeguards Rule guidance
S11	FTC breach reporting amendment to the Safeguards Rule
S12	NIST AI Risk Management Framework
S13	Maybe Finance open-source repository and firsthand product postmortem context
S14	Actual Budget official site
S15	Firefly III official site

s1 <https://www.quicken.com/products/business-personal/>

s2 <https://www.quicken.com/products/classic-business-personal/>

s3 <https://help.monarch.com/>

s4 <https://digits.com/>

s5 <https://quickbooks.intuit.com/ai-accounting/>

s6 <https://plaid.com/docs/>

s7 <https://www.sec.gov/newsroom/press-releases/2017-52>

s8 <https://www.sec.gov/>

s9 <https://www.consumerfinance.gov/compliance/compliance-resources/other-applicable-requirements/personal-financial-data-rights/>

s10 <https://www.ftc.gov/business-guidance/privacy-security/gramm-leach-bliley-act>

s11 <https://www.ftc.gov/>

s12 <https://www.nist.gov/itl/ai-risk-management-framework>

s13 <https://github.com/maybe-finance/maybe>

s14 <https://actualbudget.org/>

s15 <https://www.firefly-iii.org/>

Uploaded KEEL archive reviewed

- 00-README-overview.md
- 01-era-deep-dive.md
- 02-monarch-deep-dive.md
- 03-copilot-deep-dive.md
- 04-ramp-brex-corporate-spend.md
- 05-adjacent-players.md
- 06-feature-demand-and-gaps.md
- 07-master-feature-taxonomy.md
- 08-quicken-family-deep-dive.md
- 09-KEEL-BUILD-PLAN.md
- 10-KEEL-TECH-SPEC.md
- 11-KEEL-design-concept.html
- 12-KEEL-GAP-AUDIT.md
- 13-KEEL-ADDENDUM-v1.1.md
- 14-KEEL-FINAL-AUDIT.md
- CLAUDE.md
- 15-KEEL-BACKEND-CANONICALIZATION-v2.1.md
- 16-KEEL-v2.1-CHANGELOG.md